

African Medical Research Council also adopted the system by the end of 2006. The growth of OpenMRS has been both consistent and phenomenal. It is now used in Lesotho, Zimbabwe, Mozambique, Uganda, Tanzania, Haiti, India, China, the United States, Pakistan, the Philippines, Chile, Nicaragua, Japan, Peru, Haiti and many other countries.

Today, OpenMRS is supported by a host of renowned organisations, including the Canadian International Development Research Centre (IDRC), the Center for Disease Control (CDC), the Fogarty International Centre, Google, The Millennium Villages Project of the Earth Institute, Columbia University, The Rockefeller Foundation, ThoughtWorks Inc. and the World Health Organisation. The OpenMRS is built by volunteer developers from many backgrounds, including students, independent open source volunteers, and professional software engineers. They play a major role in nurturing a growing network of volunteers and implementers who are dedicated to building better medical record systems for the world.

The OpenMRS recipe for success

The OpenMRS project achieved great success due to the following reasons:

- OpenMRS is an attempt to solve a problem. It is a much needed answer to a serious healthcare crisis affecting the underdeveloped world.
- OpenMRS was launched by two industry leaders, both of whom made serious commitments to the success of the project. A FOSS project would never have been able to come so far if not for the sponsorship of such esteemed institutions.
- OpenMRS was not an attempt to reinvent the wheel. It was an attempt to redesign the wheel, based on prior experience. OpenMRS was designed on top of an extensive knowledge-base gathered by Regenrief and PIH over the course of many other EMR implementations.
- OpenMRS is the result of a fruitful collaboration between developers and end implementers. Engineers proactively seek feedback from implementers and medical practitioners. Implementers are encouraged to provide feedback, to join mailing lists and to be part of the community. Much effort is taken to learn from the limitations reported by other EMRs. Even more efforts are taken to ensure that OpenMRS does not suffer from these same restrictions.

What attracts the developing world to OpenMRS?

A wide majority of existing EMR solutions were planned and built for the developed world. Developers of these systems inadvertently ended up making the following assumptions during design and development:

- That the system would never be constrained by limited resources.
- That the system would be used by well-trained and qualified staff.
- The system would be maintained by capable developers.
- Implementers would be well-funded and in a position to

face any eventuality.

In most cases, a product designed for the developed world is 'handed down' to the third world almost as an afterthought. The general assumption is that a product that works well in the developed world would work as well in the developing world. However, the ground reality is very different. Unfortunately, many third world implementations lack a few (if not all) of the above criteria. Also, healthcare centres of the developing world play a more important role than those of developed nations. After all, citizens of the developed world enjoy a wide range of healthcare facilities. Therefore, a fault in a single or a few hospitals would not majorly impact patients in wealthy nations. However, implementations in the developing world are often few in number (yet cater to the needs of a very large number of people) and are the only safeguards against any epidemic disaster; hence, the need to ensure that they function well.

However, OpenMRS was designed and built specifically for the developing world, and this is why it has been such a success. Let us consider the key features that have led to the success of OpenMRS:

- First, its modular architecture allows implementers to plug in and use the modules within minutes. Adding and removing modules is so easy that it can be done with no development skills at all.
- OpenMRS supports internationalisation. The application already supports six languages. Adding support for your own language is as easy as adding a single message file to the system.
- Other features include flexible reporting tools, adherence to proper medical standards and convenient data entry/export features. These features ensure that maintaining or using an OpenMRS implementation requires a minimal amount of skills.

OpenMRS has also stepped in to tackle the lack of resources in a multitude of other ways. It has developed a synchronisation module that allows systems to synchronise their data offline. This tool is a godsend for implementers who suffer because of slow Internet speeds.

Another issue affecting the developing world is the limited number of trained professionals, and the distance between patients and hospital sites. OpenMRS' solution to this problem was to support SMS integration, so that community health workers can interact with OpenMRS via their cell phones. These features have contributed greatly to making OpenMRS a more attractive solution for the developing world.

The OpenMRS community: Attracting and keeping volunteers

Despite enjoying the support of several donor organisations, OpenMRS still relies heavily on volunteer contributors. The OpenMRS developer community is extensive, and is quite active in improving the OpenMRS core, supporting implementations and building external modules. The project has been able to attract (and keep) such a large number of volunteers largely due to the professionalism, enthusiasm and fair treatment meted out to all.